

①

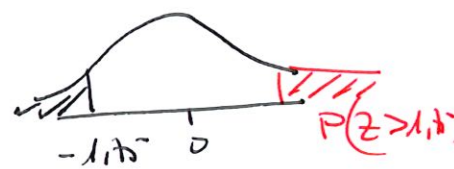
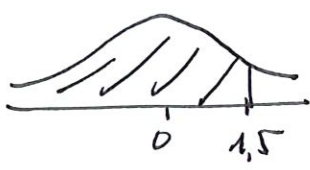
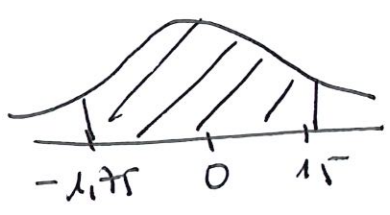
29. 439.01 $X \sim N(10, 2)$ Bauketo Normab.
 $N(\mu, \sigma)$

a) 6,5 eta 13 ordu bitartean.

$$P(6,5 < X < 13) = P\left(\frac{6,5 - 10}{2} < Z < \frac{13 - 10}{2}\right) =$$

$$Z = \frac{X - \mu}{\sigma}$$

$$P(-1,75 < Z < 1,5) = P(Z < 1,5) - P(Z < -1,75)$$



$$= P(Z < 1,5) - (1 - P(Z < 1,75)) =$$

$$= 0,9332 - (1 - 0,9599) = 0,8931 \Rightarrow \boxed{78,31}$$

b) 7 ordu baino gutxiagoan.

$$P(X < 7) = P\left[Z < \frac{7 - 10}{2}\right] = P(Z < -1,5)$$

$X \sim N(10, 2)$

$Z \sim N(0, 1)$

$$Z = \frac{X - \mu}{\sigma}$$

tipifikotua.



$$= P(Z > 1,5) = 1 - P(Z < 1,5) = 1 - 0,9332 =$$

$$= 0,0668 \rightarrow \boxed{7,668}$$



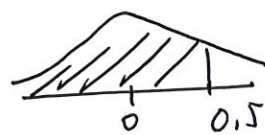
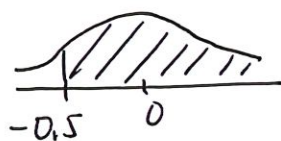
(2)

c) 9 ordu baino gehiagoan.

$$P[X > 9] = P\left[Z > \frac{9-10}{2}\right] = P[Z > -0,5] = P[Z < 0,5]$$

$$= 0,6915$$

$$\rightarrow \boxed{7.69,15}$$

**30** $X \sim N(26, 6) \rightarrow N(\mu, \sigma)$ a) Perbua bat eta 23 eta 30 tarteko GMI

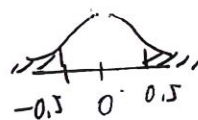
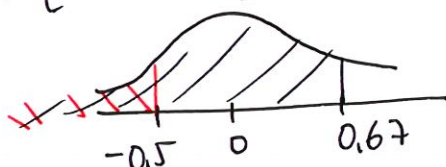
$$P[23 < X < 30] = P\left[\frac{23-26}{6} < Z < \frac{30-26}{6}\right] =$$

$$\text{Tipirikatu}$$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z \sim N(0,1)$$

$$= P[-0,5 < Z < 0,67] = P[Z < 0,67] - P[Z < -0,5] =$$



$$= P[Z < 0,67] - P[Z > 0,5] =$$

$$= P[Z < 0,67] - [1 - P[Z < 0,5]] =$$

$$= 0,7486 - (1 - 0,6915) = \boxed{0,4401} \Rightarrow \underline{7.44,01}$$

23 eta 30 tarteko GMI izateko prob. 0,4401.

b) 35 etik gorako GMI izateak $\rightarrow$ elurkeko

$$P[X > 35] = P\left[Z > \frac{36-26}{6}\right] = P[Z > 1,5] = 1 - P[Z < 1,5]$$

$$= 1 - 0,9332 = 0,0668 \rightarrow$$

$$\boxed{7.6,68}$$

Perbua ebazten elurkeko 7.6,68

31.

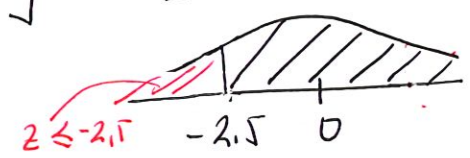
Banaketza Normale $X \sim N(1500, 200)$
 $\mu = 1500$ $\sigma = 200$

a) Zonau foku bat hartu eta gutxienez 1000 ordut
emateko arfio

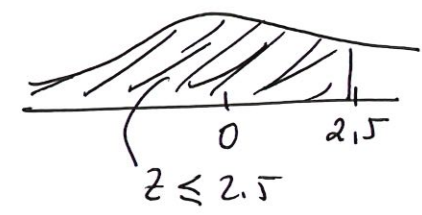
1000 ordut edo gehiago.

$$P[X \geq 1000] = P\left[Z \geq \frac{1000 - 1500}{200}\right] = P[Z \geq -2.5]$$

Tipifikatu
 $Z = \frac{X - \mu}{\sigma}$



$$= P[Z \leq 2.5] = \boxed{0.9938}$$



b) 2000 foku erosi badira, zeubatek espero
 behar duzu emotea gutxienez 1000 ordut?

$$0.9938 \cdot 2000 = 1987.6.$$

Beraz 1987 fokuk espero duzu emotea 1000 ordut
 gutxienez.

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Motor jakin
 batek irauten
 duena (urtak)

$X \sim N(10, 2)$

$\mu = 10$
 $\sigma = 2$

13 urtez ondo funtzionatzen debe.

Motorren zer ehunekok uste behar duzu et
dute garantia beteko?

Ez dute beteko 13 urtez et badeba ondo
 funtzionatzen. $\rightarrow P[X < 13]$

$$P[X < 13] = P\left[Z < \frac{13-10}{2}\right] = P[Z < 1.5] = 0.9332 \quad (4)$$

Tipifikazio $Z = \frac{X-M}{\sigma}$
 $X \sim N(10, 2) \rightarrow Z \sim N(0, 1)$

$$\Rightarrow \boxed{0.9332}$$

0.9332 -ak usteko eta dauka garantia beteko.

33

Azken nota $\rightarrow X \sim N(5.8; 2)$

$\mu = 5.8$
 $\sigma = 2$

Notarik onenak, goi-mailako ikasketak $\rightarrow X.20$

Zer da gutxienez lortu beharreko nota, goi-mailak ikasketak egiteko?

$a \equiv$ goi-mailako ikasketarako nota



$$P[X \geq a] = 0.2$$

$$P[X < a] = 0.8$$

$$P[X < a] = 0.8$$

Tipifikatuz
 $Z = \frac{X-M}{\sigma}$

$$P\left[Z < \frac{X-5.8}{2}\right] = 0.8$$

	0.04
0.8	0.8

$$\Rightarrow P[Z < 0.84] = 0.8$$

$$\frac{X-5.8}{2} = 0.84 \Rightarrow \boxed{X = 7.48}$$

7.48 da gutxienezko nota goi-mailako zikloetan sartzeko.

34

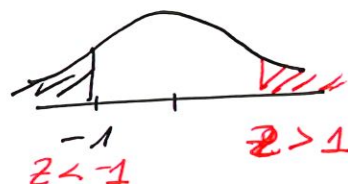
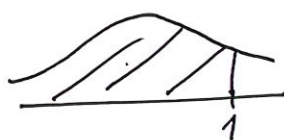
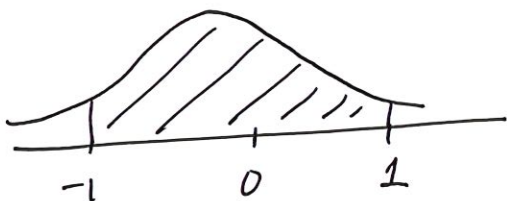
Adimen kolientea $X \sim N(100, 10)$
AK. $N(\mu, \sigma)$

5

a) zer ehuneko duen 90 eta 110 birtetako AK.

$$P[90 < X < 110] = P\left[\frac{90-100}{10} < Z < \frac{110-100}{10}\right] =$$

$$= P[-1 < Z < 1] = P[Z < 1] - P[Z < -1]$$



$$= P[Z < 1] - (1 - P(Z < 1)) = 0,8413 - (1 - 0,8413) =$$



$$= 0,6826$$

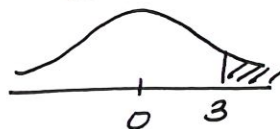
$$\Rightarrow \boxed{\% 68,26}$$

%68,26-ak dauko 90 eta 110 birtetako AK

b) 130 etik gora $\rightarrow$ SUPERDOTANTZAT zein da ehuneko?

$$P[X \geq 130] = P\left[Z \geq \frac{130-100}{10}\right] = P[Z \geq 3] =$$

$$= 1 - P[Z \leq 3] = 1 - 0,9987 =$$



$$= 0,0013. \rightarrow \boxed{\% 0,13}$$

c) %70-ek balio jakin batetik beherako A.K. zein da balio?

$$P[X \leq a] = 0,7.$$

$$P\left[Z \leq \frac{a-100}{10}\right] = 0,7$$

Taulatik:

$$\frac{a-100}{10} = 0,525$$

$$a = \underline{\underline{105,25}}$$

0,02 0,03

0,6950 0,7019

Populazioko %70-ek, 105,25 baino beherako AK daukete.

35. $N(\mu, \sigma)$

Saioak 172 cm.
Bejõu 167 cm

Normalk ⑥
1,4
0,4

a) 2e garaiera du Eiderrek, tipifikatua -1 bado!

$$z = \frac{x - \mu}{\sigma}$$

$$\left. \begin{array}{l} \text{Saioa} \quad \frac{172 - \mu}{\sigma} = 1,4 \\ \text{Bejõu} \quad \frac{167 - \mu}{\sigma} = 0,4 \end{array} \right\}$$

$$\begin{array}{r} 172 - \mu = 1,4\sigma \\ - 167 - \mu = 0,4\sigma \end{array}$$

$$\boxed{5 = \sigma} \rightarrow \boxed{\mu = 165}$$

Eiderren kosuou $\frac{x - 165}{5} = -1 \rightarrow \boxed{x = 160 \text{ cm}}$

b) zein da Auhazeren goioko tipifikatua, errealitate,
165 cm

$$\frac{165 - 165}{5} = 0. \Rightarrow \underline{\underline{0}} \text{ da goiiera tipifikatua}$$

36. Piezondimetroa $x \sim N(45, \sigma)$
(mm)

a) zein da σ jokianda $P[X > 50] = 0,0062$.

$$P[X > 50] = 0,0062.$$

$$P[X \leq 50] = 1 - 0,0062 = 0,9938.$$

0,00
2,5 $\boxed{0,9938}$

$$P[2 \leq 2,5] = 0,9938$$

$$2,5 = \frac{50 - 45}{\sigma} \Rightarrow \boxed{\sigma = 2}$$

$$2,5 \cdot \sigma = 5$$

b) 820 prezės atšertutė, ženkstok
išaujo dūtė 39,7 etė 43,5 mm intortek diometra.

Leheueujo prezė bat atšertutė žėu dō prob
atšertuendōju.

$$X \sim N(45, 2)$$

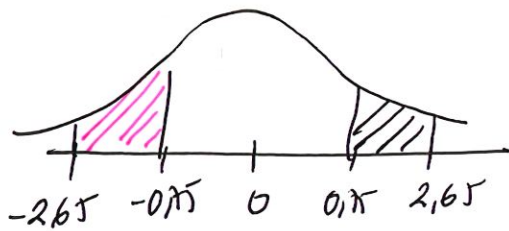
$$P[39,7 < x < 43,5] =$$

$$= P\left[\frac{39,7 - 45}{2} < z < \frac{43,5 - 45}{2}\right] = P[-2,65 < z < -0,75]$$

$$= P[0,75 < z < 2,65] =$$

$$= P[z < 2,65] - P[z < 0,75]$$

$$= 0,9960 - 0,7735 = \boxed{0,2225}$$



820 prezė badiŗa

$$820 \cdot 0,2225 = 182,45 \approx \boxed{182 \text{ prezės}}$$